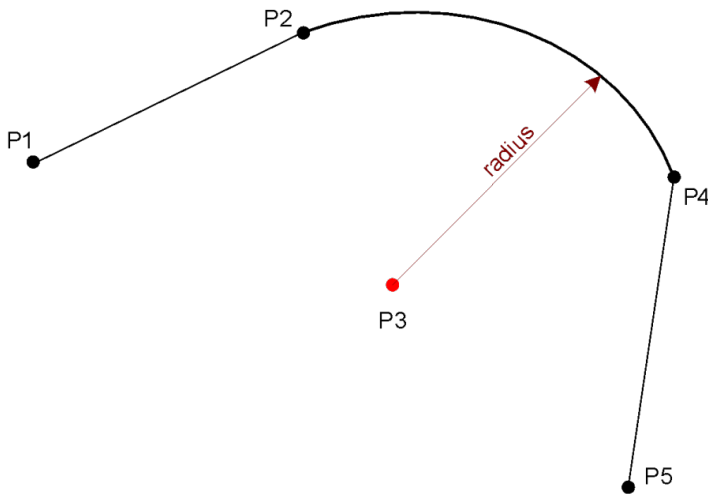


Mapping for ArcByCenterPoint

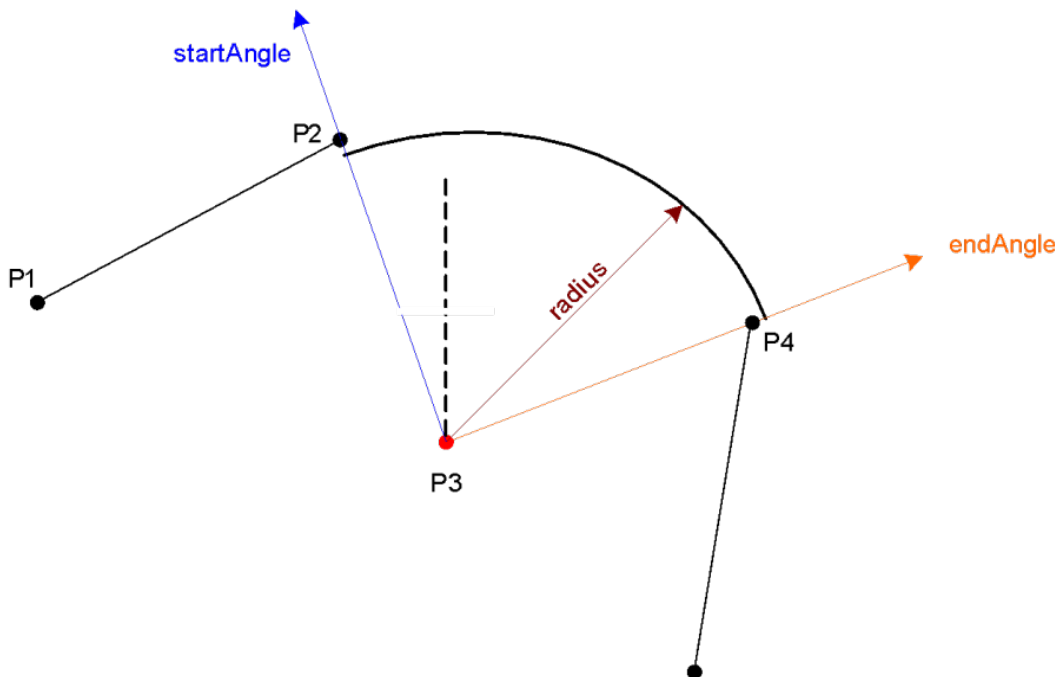
This page indicates how to calculate the start/end angles of an [ArcByCenterPoint](#) when the information provided by the (official) source is in the form of start/end positions, centre and radius. The situation presented in the figure below exemplifies a part of an Airspace boundary that is a sequence of:

- a straight segment (P1-P2)
- followed by a clockwise arc from P2 to P4 with the centre in P3
- followed again by a straight line to P5.

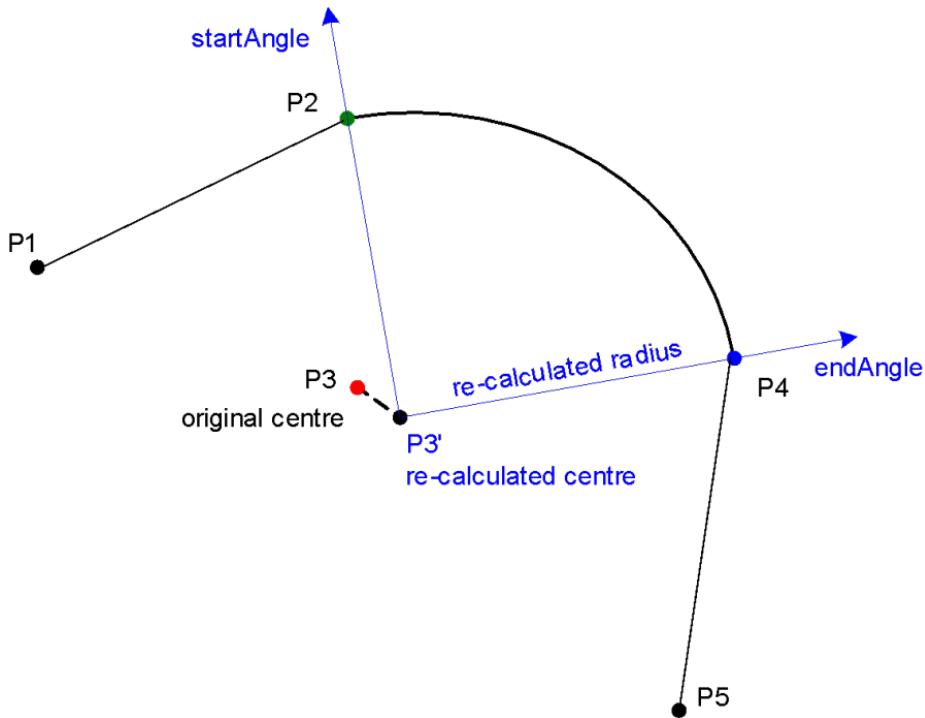


Sometimes, trying to draw the arc using the exact values provided in the official sources for the centre point, radius and positions P2 and P4 does not work. The declared radius may be significantly different* from the calculated distances from the centre P3 to points P2 or P4. An objective of the GML encoding is to preserve as much as possible the original (official) information.

Frequently, the centre of the arc corresponds to the position of a physical object (such as a Navaid) and the radius value is meant to prevent airspace users from crossing the airspace boundary. Therefore, it is recommended that the original centre and radius data is used directly for the [ArcByCenterPoint](#) in the AIXM/GML encoding. The [startAngle](#) shall be calculated using the vector P3-P2 and the [endAngle](#) shall be calculated using the vector P3-P4. The resulting arc is represented in the figure below. This shows that it is possible that the resulting GML arc does not align at points P2 and P4, because of the imprecision of the original centre and/or radius data.



If the potential misalignment of the arc (as indicated in the figure above) is a problem for an application, then it is suggested that, for example, the original centre and radius data are replaced with calculated values, by that application, as indicated in the figure below:



The objective is to find a point that is at equal geodesic distance from the start P2 and end P4 points. This can be done with an algorithm that iteratively tries to find the point of which the distance to both P2 and P4 is the same and which is as close as possible to the original center. However, if the difference between the original distances P3-P2 and P3-P4 is higher than 1%, then the application shall raise an error message and abort the calculation. This rule was also specified in previous AIXM versions (4.5) and it did prevent the provision of geometrically inconsistent arc data.

The calculated point (P3') can then be used as the corrected center point. The corrected radius is the distance from the corrected center P3' to P4 (because of the algorithm applied, this distance is also the distance to P2). The start/end angles are the bearings from the corrected center P3' to P1/P2.

There might be situations where the centre P3 is an Aerodrome Reference Point (ARP) or a Navaid. In this situation, it might be more appropriate to re-calculate the points P2 and P4 in order to correctly close the surface. This could be problematic when P2 or P4 are also ends of arcs. The best solution depends on the intended use of the data, therefore this decision is left to application developers.

Footnotes

* For example, in the European AIS Database, a difference of more than 1% is considered significant and raised as an error, that requires corrective action.